MACHINE LEARNING · POLITICAL SCIENCE · SURVEY DATA

Predicting Attitudes Toward Democracy

A Machine Learning Case Study

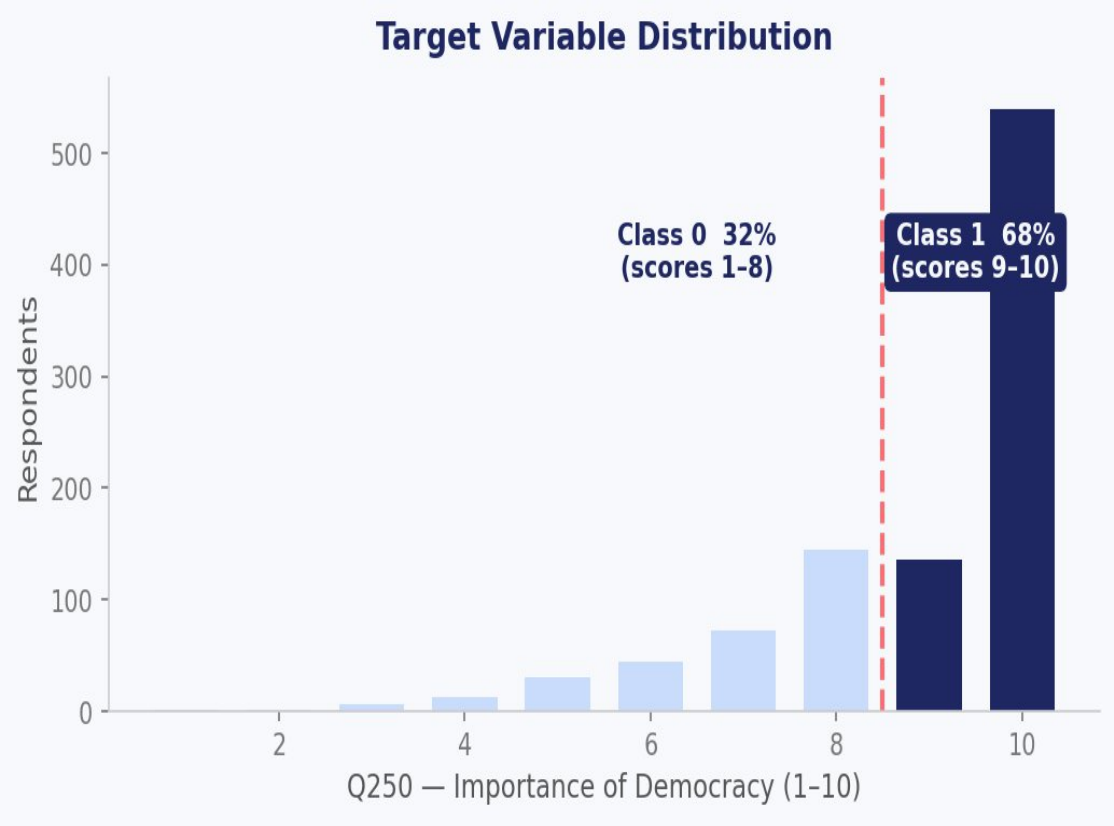
World Values Survey · Wave 7 · Argentina

CORE QUESTION

What distinguishes people who strongly value democracy from those who don't - and can we predict it from other survey responses?

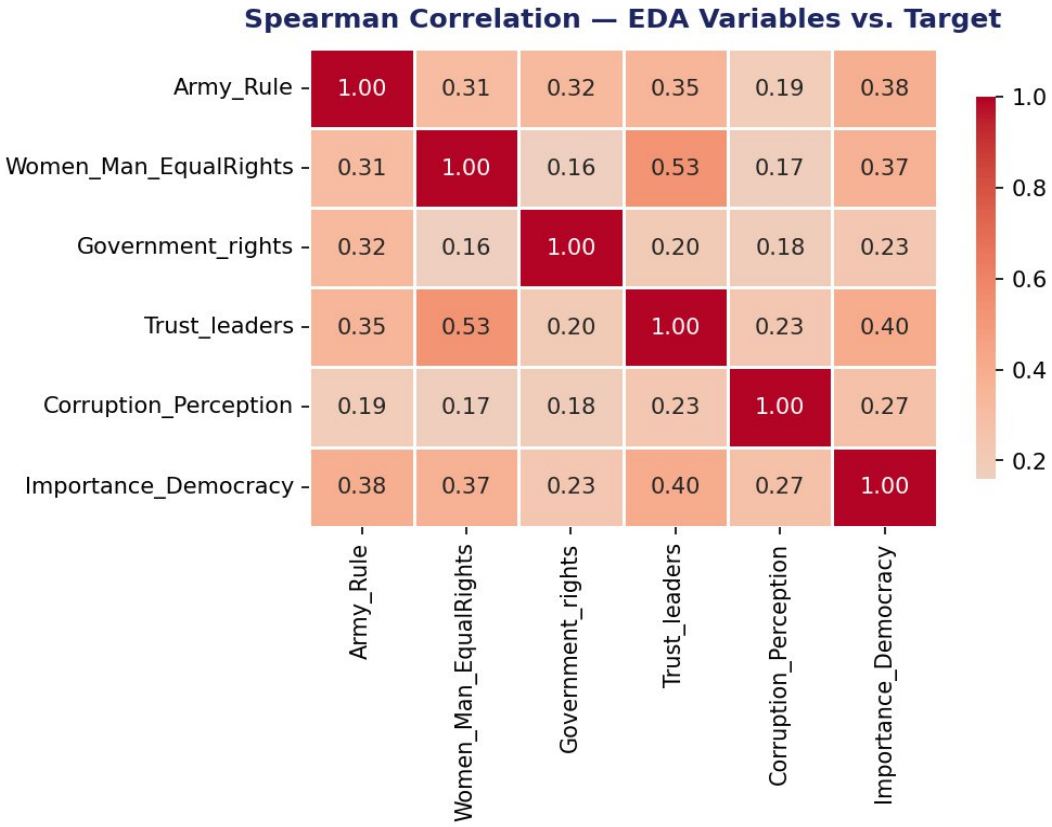
DATASET	SAMPLE	TARGET
World Values Survey Wave 7 — Argentina	1,003 respondents ~395 variables	Q250: Importance of Democracy (1–10)

TARGET VARIABLE



Binarization Strategy

- Q250 is highly skewed (most respondents cluster at 9-10).
- Threshold applied at score > 8 to create a meaningful binary split
- Class 0 (scores 1–8): 32% - lower democratic valuation -.
- Class 1 (scores 9–10): 68% - higher democratic valuation -.
- Class imbalance (~2:1) addressed by optimizing for F1-score, not accuracy



Key Patterns

Army_Rule (0.38)

Strongest EDA signal - rejecting military rule aligns with higher democratic valuation -.

Trust in Elections (0.40)

Institutional trust and democratic valuation move closely together

Gender Equality (0.37)

Egalitarian attitudes correlate with democratic values, and also with institutional trust (0.53)

Corruption (0.27)

Weak signal: people value democracy even when they perceive high corruption

MODELING PIPELINE

1
Preprocess

Negative values → NaN
Mode imputation
Standard scaling

2
Select

SelectKBest
F-test vs binary target
k = 6 features

3
Compare

Logistic Regression
Random Forest
XGBoost · CatBoost

4
Tune

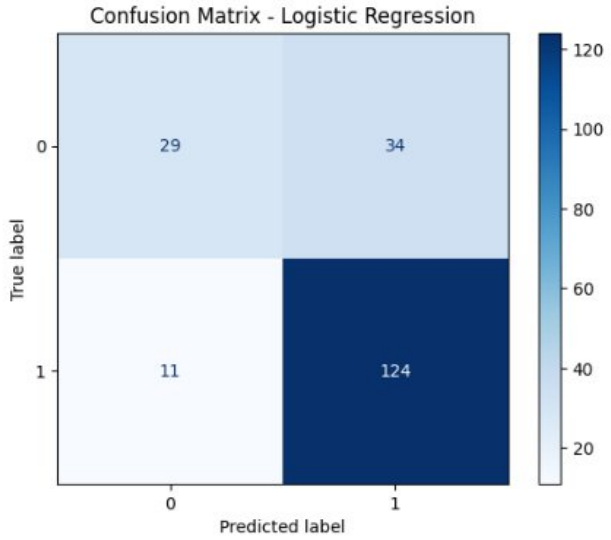
GridSearchCV
Stratified 5-fold CV
Optimize F1-score

All preprocessing steps are performed inside the sklearn Pipeline — no data leakage across cross-validation folds.

MODEL RESULTS

WINNER: LOGISTIC REGRESSION

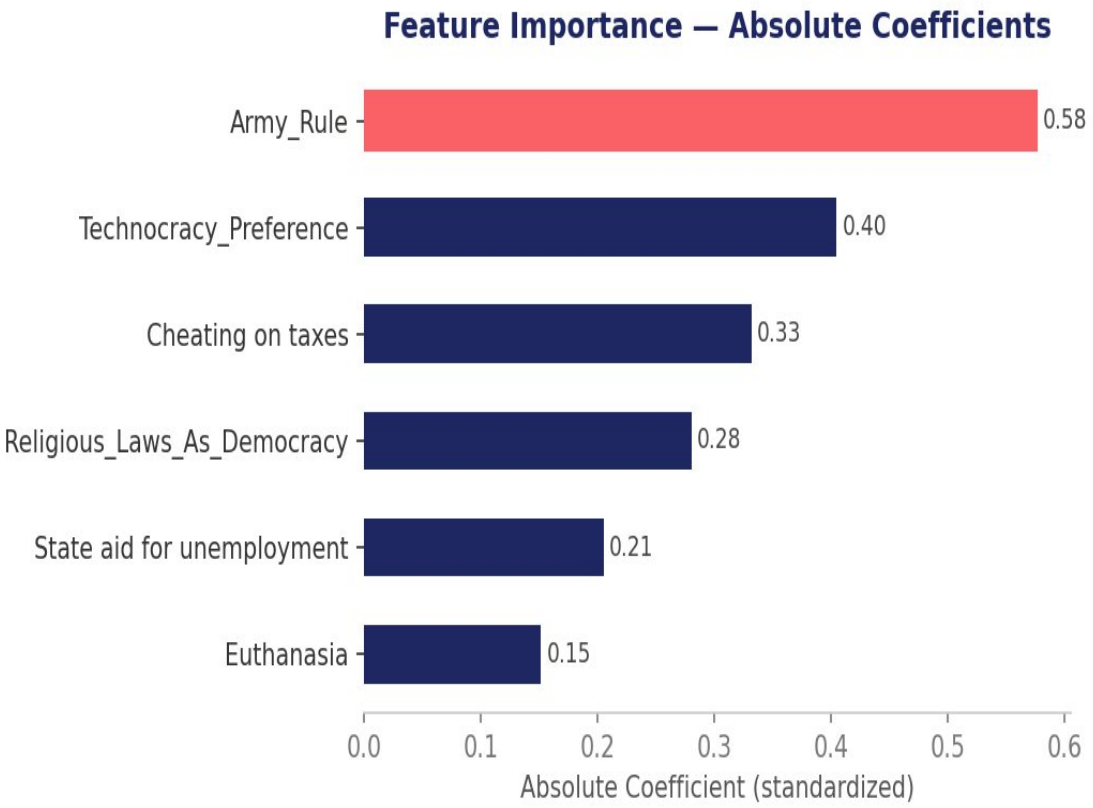
77%	0.85	0.56	0.792
Accuracy	F1 - Class 1	F1 - Class 0	ROC-AUC



RESULTS

- Logistic Regression with C=10 and k=6 features achieved the best cross-validated F1.
- Expected: regularized linear models often outperform ensembles on small datasets with threshold-based targets.
- F1 optimized over accuracy: deliberate choice given the 2:1 class imbalance.
- ROC-AUC of 0.792 confirms meaningful discrimination above a naive baseline.
- 11 false negatives: respondents who genuinely scored 9-10 but were predicted as low-valuation.

FEATURE IMPORTANCE



What the model uses

6 features selected

- SelectKBest with F-test on binarized target (ranking differs from Spearman used in EDA).

Army_Rule dominates

- Largest absolute coefficient (0.57), consistent with EDA findings.

Diverse domains

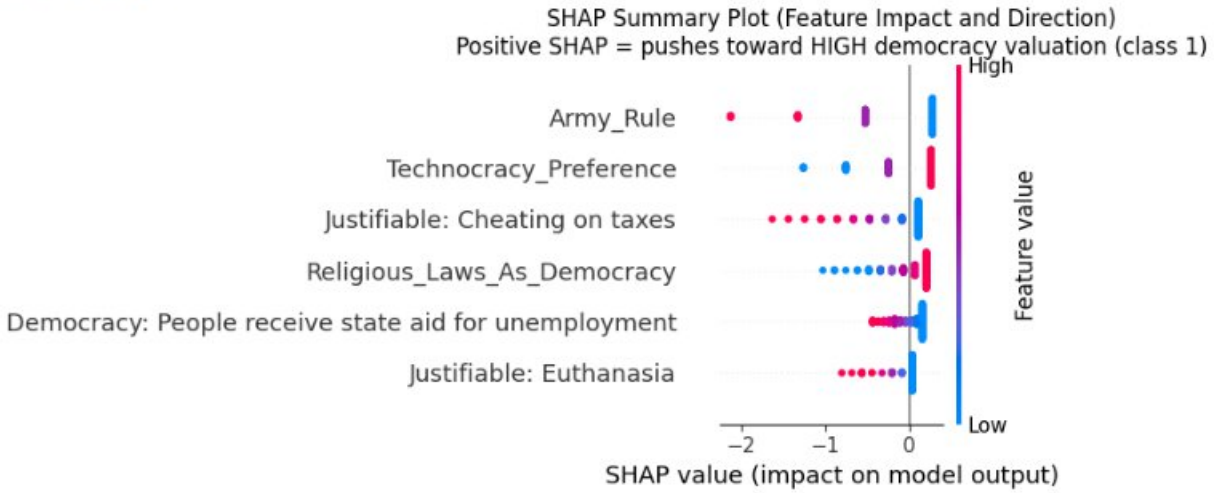
- Political system attitudes + conceptions of democracy + moral attitudes

Coefficients ≠ causal

- Magnitude shows importance; direction reflects multivariate partial effects (see next slide)

EXPLAINABILITY — SHAP DIRECTIONALITY

SHAP Beeswarm Plot



- Each dot is one respondent in the test set
- Position on X axis = magnitude and direction of impact on the prediction
- Color = feature value (red = high, blue = low)
- Positive SHAP = pushes toward Class 1 (high democratic valuation)

Note on suppression

Some directions differ from EDA correlations. When correlated features enter together, the model estimates partial effects. Individual signs can flip. This is statistical suppression, not a modeling error.

LIMITATIONS

01

No causal inference

Cross-sectional observational data. The model identifies associations, not mechanisms.

02

Single-country sample

Argentina only (WVS Wave 7, n ≈ 989). Findings may not generalize across cultural contexts

03

Self-report bias

Social desirability effects are plausible in items about democratic values.

04

Threshold sensitivity

Binarization at score > 8 is a modeling choice. A different threshold produces a different model.

05

Feature stability

Data-driven selection (SelectKBest) may retain variables correlated with the target for sample-specific reasons

06

Coefficient interpretability

Statistical suppression affects individual coefficient signs. SHAP reflects partial effects, not univariate relationships

DIVE DEEPER

Full analysis, code, and interpretation available in the notebook.

github.com/tomasallemand/wvs-democracy-attitudes-ml

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Haerpfer et al. (2022) · doi:10.14281/18241.20